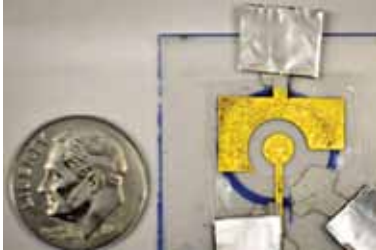




Ultra-low-cost sensor detects diseases like cancer, HIV, and flu

MIT's (Cambridge, Massachusetts) ultra-low-cost sensor uses DNA-coated gold electrodes and CRISPR-Cas12 to



The electrochemical sensors developed in Ariel Furst's lab consist of DNA adhered to an inexpensive gold leaf electrode, which is laminated onto a sheet of plastic (Credit: Courtesy of the researchers; edited by MIT News)

detect diseases like cancer, HIV, and flu through an electrochemical signal, just like a glucose meter. A polyvinyl alcohol (PVA) layer shields DNA from heat and degradation, extending shelf life to two months, even in high-temperature settings, while costing under

a cent. Pre-stable, refrigeration-free, and compatible with urine, saliva, or nasal samples, it is perfect for low-resource clinics and rapid disease detection worldwide.

Smart dry adhesive for next-gen smart micro-assembly

Researchers at POSTECH (Pohang, South Korea) have developed a smart dry adhesive that enables the precise pick-and-place of ultra-small objects such as micro-LEDs, without residue or complexity. Led by Prof. Seok Kim, the team embedded nano-tips in shape-memory polymers that switch between sticky and non-sticky states with the application of heat. This reversible grip tackles the 'adhesion paradox' head-on, handling a wide range of materials, from electronics to fabric. It is a clean, residue-free, high-force solution poised to transform micro-assembly in displays, semiconductors, and next-gen smart manufacturing.

Robotic probe to accelerate development of next-gen materials

Researchers at MIT (Cambridge, Massachusetts) have developed a fully autonomous robotic system that accelerates the discovery of advanced semiconductors for electronics and solar technology. The robot, guided by a self-supervised AI, pinpoints optimal probe points and collects over 125 photoconductance measurements per hour, which are far faster and more accurate than manual methods. With computer vision and smart path planning, it handles diverse, snowflake-like samples with ease. Backed by industry giants, this breakthrough could accelerate the development of next-generation materials for sustainable technology.

Thin films compatible with standard silicon tech, ideal for optical computing

The Tokyo Institute of Science has engineered (Al,Ga,Sc)N thin films with record-high scandium levels, unlocking ultra-low-power ferroelectric memory and 6G filters. By tweaking deposition parameters using reactive magnetron sputtering, the team achieved polarisation switching at just 1.8MV/cm, one of the lowest values ever reported. These films also exhibit enhanced piezoelectric and optoelectric properties, making them ideal for optical computing. Crucially, they remain compatible with standard silicon tech, making this a scalable solution for next-gen electronics and communication systems.

Robots use sound waves to move objects

Researchers at University College London have developed a swarm robotic system that levitates and moves objects using only sound waves, eliminating the need for physical



These robots work together to levitate objects with sound waves (Credit: N. Kemsaram et al)

contact. Inspired by ants, each robot uses ultrasonic transducers to create precise acoustic pressure fields. Multiple robots can synchronise to lift larger objects cooperatively with microsecond precision. This breakthrough enables ultra-clean, hands-free material handling for future applications in micro-assembly, sterile environments, and medical labs, where even the lightest touch could be too much.

Ultra-intense laser pulses captured in real time, in one shot

Researchers at Oxford, LMU Munich, and the Max Planck Institute have unveiled RAVEN, a system that captures ultra-intense laser pulses in real time, in just one shot. Unlike previous multi-shot methods, RAVEN offers instant, full 3D mapping of a pulse's shape, timing, and polarisation, enabling real-time correction of distortions like spatio-temporal coupling. This significantly enhances precision and efficiency in high-energy experiments, including fusion energy, particle acceleration, and quantum electrodynamics, paving the way for next-generation laser-driven research and clean energy innovations.